

Pranay S B

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3rd-year Electrical Engineering undergraduate. Open-source contributor to NVIDIA, OpenTelemetry, Voxel51, and Roboflow. Leading a government-funded edge vision prototype.

Education

Indian Institute of Technology Bhilai

2024 – 2028

B.Tech in Electrical Engineering

CGPA: 8.51/10

Experience

Project Lead, AI Vision Systems — MeitY GENESIS Grant Project

December 2025 – Present

Entrepreneur-in-Residence (EiR) Program, IIT Bhilai

- Selected for the competitive MeitY GENESIS EiR program and secured a **Rs 10 lakh (~USD 12,000) competitive innovation grant** to prototype AI-driven vision sensing systems for smart infrastructure applications.
- Lead the technical development of the prototype — architecture, backend system design, AI inference pipelines, and Dockerized deployment — building low-latency vision intelligence modules for constrained edge and cloud hardware.

AI/ML Research Intern

May 2025 – January 2026

Visual Intelligence and Learning Lab, IIT Hyderabad

- Reproduced and benchmarked DaViT and GenAD on real-world sensor datasets, documenting generalization under distribution shift; implemented 3D Gaussian Splatting for dynamic object removal.
- Built a modular BLIP-based VQA pipeline on BDD100K with custom annotation tooling and evaluation harnesses. [\[Live Demo & Model\]](#)

Open Source Contributions

[pranaysb.me/open-source](#)

- 12 merged pull requests across 6 organizations** — NVIDIA NeMo-Agent-Toolkit, OpenTelemetry Python (CNCF), Voxel51/FiftyOne, embeddings-benchmark/mteb, Roboflow/trackers, and Kornia. Of 15 defects reported, authored the fix for 12.
- NVIDIA NeMo-Agent-Toolkit** (3 merged) — eliminated severe memory leaks from unbounded `@lru_cache` on instance and static methods in core utilities and `DiscoveryMetadata`; fixed CLI `--override` flags being silently ignored by the finetuning runtime.
- OpenTelemetry Python** — fixed a data-integrity defect where `Context` objects were mutable in-place via inherited `dict` methods, violating the documented immutability contract.
- Roboflow/trackers** (2 merged) — fixed an unhandled `cv2.error` crash in camera-motion compensation and a zero-division in the `xcycsr_to_xyxy` coordinate converter; separately reported a Zip Slip path-traversal vulnerability, patched directly by maintainers; credited as a contributor in the project's latest release notes.
- Voxel51/FiftyOne** (2 merged) — added visible-bbox and visibility-ratio fields to Cityscapes person annotations for occlusion-aware evaluation; added validation warning for reserved field names.

Projects

API Forge AI — Agentic SDK Generation Platform

[Live](#) [GitHub](#)

- Built a multi-agent LangGraph platform (Planner, Validator, Diagnoser, Coder, Executor) that converts OpenAPI specs into production-ready SDKs with self-healing code correction; stack: FastAPI, Next.js, PostgreSQL, Docker.

IPL Verse — Full-Stack Daily IPL Gaming Platform

[Live](#) [GitHub](#)

- Built and shipped a public, full-stack daily gaming platform for IPL fans (5 game modes including multiplayer Arena battles) with Next.js, Supabase/PostgreSQL, authentication, and real-time multiplayer across 7-round matches.
- Engineered an AI content pipeline (Groq API) generating and curating 3,600+ trivia questions with Sentence Transformer deduplication cutting redundancy by 70%+.

Technical Skills

Programming Languages: Python, C++, TypeScript, JavaScript, Ruby

Backend & Databases: Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, MongoDB, Supabase

AI / ML: PyTorch, OpenCV, HuggingFace Transformers, LangGraph, Multi-Agent Systems, LLMs, RAG, Semantic Search

Developer Tools: Git, GitHub, GitHub Actions, Docker, Linux, OpenTelemetry, CI/CD

Achievements & Positions of Responsibility

- 1st Place, Build with Gemma 4 (AI Durg)** — built a voice-first AI assistant using Google's Gemma open-weight models, from 65 competing teams.
- 3rd Place, Pathway Hackathon** — real-time financial analytics platform built on Pathway's streaming tools.
- 2nd Place, TechSprint** (Google Developer Groups) — Flutter-based, Gemini-integrated campus app.
- AI/ML Lead, GDG & Core Member, DS&AI Club, IIT Bhilai** — led deep learning workshops and bootcamps; mentored 30+ peers.